

Expanding Clinically Relevant Health Monitoring

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CCS Concepts

• **Applied computing** → **Life and medical sciences**; • **Human-centered computing** → **Human computer interaction (HCI)**; **Ubiquitous and mobile computing**.

Keywords

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1 Problem statement

The current US healthcare system is plagued by inequities and failing health measures for large portions of the population. Blood pressure (BP) management exemplifies this problem. Hypertension increases the risk of heart failure and strokes, two leading causes of deaths in the United States. Although nearly half of all US adults meet the hypertension diagnostic criteria of $>130/80$ mmHg, an estimated one in three are unaware of their condition, an issue that is further compounded by disproportionate incidence with racial and ethnic minorities [4]. In 2020 the U.S. Surgeon General released a “Call to Action to Control Hypertension” which identified the control of hypertension as a national priority, laying out strategies to improve the awareness, treatment, and successful control of hypertension [12]. Community studies have demonstrated that early screening to increase awareness and promoting shared management of hypertension through self-measurement of BP are effective strategies for the successful control and treatment of hypertension [15]. However, the high cost of BP monitors deters ownership and substantial patient and health system impediments (transportation barriers, childcare limitations, lack of maternity leave, and others) limit adherence to in-person BP checks, particularly among racial and ethnic minority patients, perpetuating health disparities. Therefore, there is an urgent need for fully remote, cost effective, reliable BP measurement tools.

My Ph.D. work focuses on finding low-cost solutions to expanding healthcare access, with a significant focus on BP and hand grip strength (HGS). Although wearables and smartphone apps exist to

measure some physiological or health sensing, most of these applications are not clinically relevant. Common measurements such as continuous photoplethysmography (PPG), heart rate, steps, etc. are often not rigorously validated and do not directly fit into the clinically relevant metrics physicians use in hospitals. As such, my work focuses on developing more clinically relevant measurements that can be easily through the billions of smartphones across the world.

My proposed thesis topic focuses on two smartphone measurements: BP and HGS. These measurement of BP and HGS are similar because both measurements require the user to apply force to the smartphone. The BP measurement is performed at the fingertip and requires the user to apply force onto the digital artery. The HGS measurement requires users to squeeze the smartphone with their hand. HGS is important as a measure of fragility in clinical settings, which is particularly relevant for predicting surgical outcomes and determining surgical readiness. Both these metrics rely on a novel technique of measuring applied force on a smartphone coined “Vibrometric Force Estimation.” This technique involves vibrating the smartphone and measuring how much the vibration dampens as a result of the applied force.

2 Related Work

This work represents the first HGS and absolute BP measurement techniques on a smartphone without special components or attachments. To the author’s knowledge there is no prior work on smartphone HGS. The most relevant work includes attempts to use the smartphone as a scale or usability research investigating how to measure hand positioning and gestures [7, 17]. Most prior research into smartphone and smartwatch BP measurement perform relative BP measurements using Pulse Transit Time (PTT) or Pulse Wave Analysis (PWA), but these techniques are fundamentally different. PTT and PWA only provide relative change in BP. These measures of relative change assume an individual has access to a BP cuff (or another absolute BP measurement) [5, 6, 8–11, 13, 14]. Prior works on absolute smartphone BP measurement require custom built attachments to measure force [1, 16]. One prior work exists that functions only on a specific smartphone model with a built in force sensor [2]. However, these prior technique provide important findings and algorithm development on the oscillometric finger pressing method of estimating blood pressure. Our proposed method will be the first smartphone BP measurement that requires no attachment and functions on nearly any smartphone model with a vibration motor, camera, and IMU.

2.1 Prior Research

My prior research in my PhD includes projects to measure BP, ensure measurement consistency across different smartphones, and



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deploy smartphone pupillometry measurements to users in at-home conditions.

A previous smartphone-based BP measurement system included an ultra low cost attachment. The attachment contained a plastic clip with a spring-loaded mechanism to enable smartphones to use the LED flash and camera to measure BP. This technique focuses on utilizing smartphone components found on nearly all smartphones and relies on the same oscillometric method utilized by the novel technique discussed in this paper, but requires an attachment. Although the attachment is affordable at the cent level price, the distribution and quality assurance of the attachment becomes a difficult problem at scale. Additionally, this project brought about the idea of calibrating smartphone cameras to ensure consistency across smartphones because the project requires all phones to measure pulse similarly using the camera.

Smartphone cameras are engineered to optimize picture quality rather than measurement consistency. To achieve aesthetic pictures, smartphone default post processing involves significant nonlinearities that could compromise camera measurements. These nonlinearities can also affect pulse measurement taken via a common camera-based methodology called photoplethysmography (PPG). To account for these issues, we developed a calibration technique to ensure that PPG measurements from different smartphones remain consistent.

Aside from work in BP, my prior research involves developing and deploying novel smartphone measurements within target populations. I previously developed several iterations of pupillometry measurements, which can be clinically used to perform non-invasive cognitive tests. I further deployed this technology among older adults through a collaboration with the UC San Diego Center for Mental Health Technology to assess the impact and usability of the systems for the target population in at-home conditions.

3 Methodology

3.1 Smartphone Application

For smartphone force estimation and BP studies, we develop a custom applications to guide users and collect data. The smartphone HGS application, Dyna, provides a users with a simple experience with prompts to squeeze the smartphone. The smartphone BP application, "VibroBP" provides the user with two key visual signifiers. First, the smartphone provides an intuitive instruction of how to perform the measurement including specific finger positioning. Second, during the force measurement, the smartphone app plots the applied force in real time overlaid with a force guideline. The force guide aids the user to apply the correct amount of force during the measurement. The application additionally provides the research staff with helpful information during data collection. The PPG signal is plotted in real time before and during the measurement so the research staff can ensure data quality. At the end of the study, the full force and PPG signals are plotted within the application to provide an overview of the data.

3.2 Force Profiling

Both the proposed smartphone BP and HGS measurement methods largely depend on accurately estimating force without an attachment. Each smartphone model has its own force dampening effect

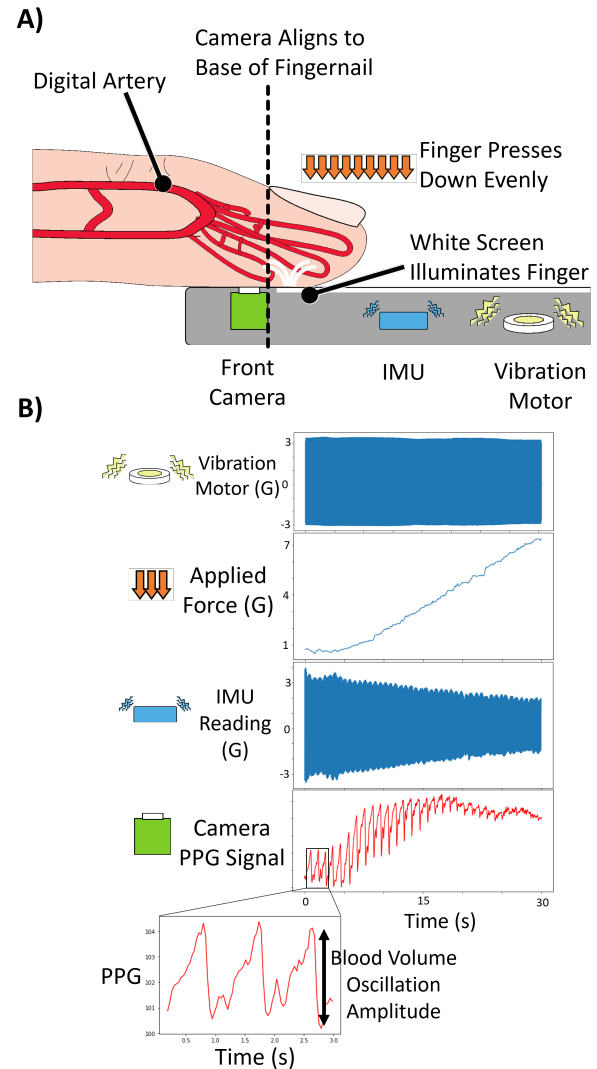


Figure 1: (A) Simplified diagram of the smartphone vibrometric force estimation, where the vibration motor drives oscillatory motion and the finger applies a force. The smartphone Inertial Measurement Unit (IMU) records the motion and the smartphone camera records PPG. (B) Applied finger force and linear y-axis acceleration of smartphone IMU are plotted vertically. As force increases, the oscillation amplitude of the y-axis acceleration decreases. The raw PPG signal is also plotted vertically to show how increased force affects the blood volume.

profile. To characterize the profile, a small-scale data collection is performed on each phone model to capture the vibration modulation at different known forces measured by a force sensitive resistor. This calibration procedure and all studies in the paper are performed on a flat wooden desktop surface.

We conducted similar profiling for three different smartphones for both Dyna and VibroBP. For VibroBP, $N=5$ participants of various hand sizes, apply pressure with their index finger onto a 0.3mm thick force sensor positioned on top of the smartphone front camera. The user applies force with their index finger in the same vertical positioning as the BP measurement. The force sensor, a calibrated, linear force sensor (SingleTact 15mm 4.5N), is placed over the smartphone front camera. The user presses their finger on the force sensor (and smartphone camera) to measure the applied force of the finger and collect smartphone data.

During a measurement, real-time readings from the ground truth force sensor are displayed on a computer screen in front of the user with a force guide. The user is instructed to use real time feedback to follow the force guide in applying a range of pressures. Like the BP measurement, the force guide instructs users to continuously increase the applied force. The study includes three repetitions of a 40 second session of applying a range of forces. These three repetitions are performed by each participant on all three smartphones for a resulting 10 minutes of raw applied force data on each phone (2 minutes per participant).

3.3 Vibrometric Force Estimation

For each smartphone, data from the force dampening effect profiling study provides IMU data during a range of applied force values. To estimate force, the IMU profile is used to train a machine learning model for each phone model. The IMU data contains three axis accelerometer data and three axis gyroscope data. Each axis of the IMU data contains high frequency signals from the vibration motor as well as noise at lower frequencies. The applied force from the finger most significantly affects the amplitude of the signal oscillation. To leverage this understanding, the rolling standard deviation of each axis of the accelerometer and gyroscope data serve as features. The features most significantly correlated to the force data are included as model inputs, while other features are excluded. With these IMU data features as inputs, a regression model (multivariate linear regression model or gradient boosting) is trained using a ten percent holdout for validation. This process is repeated for each smartphone such that a distinct force estimation model exists for each smartphone model.

3.4 Camera PPG

For the oscillometric BP measurement, we leverage the smartphone camera PPG to sense volumetric changes in blood flow at the transverse palmar arch branch of the digital artery. As the volume of blood flowing through the artery near the fingernail bed changes, the reflective properties of the tissue in the region changes. As such, the smartphone camera can record changes in blood volume by measuring the changes in the reflective properties.

In the study, the phone screen is set to a pure white background with maximal brightness and the user is instructed to place their index finger over the camera (as aided by a visual signifier). The bright white screen illuminates the finger and the camera records pixel intensity changes in the red channel as a proxy for blood volume changes. To account for skin tone and lighting variations, the smartphone performs the calibration outlined in Xuan and Barry

et. al.21 with the user's finger over the camera. This calibration protects against under- or over- saturated PPG measurements.

3.5 Oscillometry and Blood Pressure Estimation

During the smartphone BP measurement, the smartphone camera records the PPG signal while the user exerts force with the index finger on the camera. Applying force causes changes in the PPG signal to create an oscillogram shape. When the applied force is greater than the diastolic BP, the blood volume during the systolic pulse phase increases and the blood volume in the diastolic phase decreases. The total volume of blood flowing through the artery is approximately the same, but pressure on the artery causes more blood to flow through at the systolic pulse phase and less at the diastolic pulse phase. This increases the prominence of the PPG signal. When the applied force surpasses the mean arterial pressure, the blood volume begins to be limited even during the systolic phase of the pulse. As such, the systolic phase PPG prominence begins to decrease and the blood volume during the diastolic phase approaches zero. The PPG prominence continues to decrease even after the applied force surpasses the systolic BP. As the pressure increases, the signal will decrease to zero (or noise) as the blood flows through the artery during both the systolic and diastolic phase approach zero.

A plot of the pulse prominence through the range of applied forces (from less than diastolic BP to greater than systolic BP) creates a Gaussian or skew Gaussian-like shape that we refer to as the oscillogram. The oscillometric method utilized by most FDA approved BP cuffs and numerous prior academic works utilize the oscillogram shape to estimate BP. Our method of estimating BP relies on the features and findings identified in prior work.

We generate eleven features by focusing on features and concepts previously identified in prior work for oscillometric BP measurements [3]. The following features are included in the BP estimation model: the applied force at the maximum prominence value, maximum gradient, minimum gradient, applied force at the maximum gradient, applied force at the minimum gradient, the maximum value of the oscillogram, prominence at the maximum gradient, prominence at the minimum gradient, applied force at the maximal low pass filtered PPG values, and applied force at the maximum gradient of low pass filtered PPG values.

These key features of the oscillometric method are the only inputs into our BP prediction model. Note that the attributes related to exercise including heart rate and pulse shape are not included in the features provided to the model. Additionally, indirect measurements of heart rate or pulse are further avoided by interpolating all oscillograms to have the same number of points with the same force values. With the key features, we trained a least absolute shrinkage and selection operator (LASSO) regression model to predict systolic and diastolic measures of BP23. The model is trained and tested using a hold one measurement out validation.

4 Evaluation

The evaluation will focus on two key hypotheses: 1. Software-only solutions can enable smartphones to perform clinically relevant (BP and HGS) measurements. 2. Software deployments to clinical

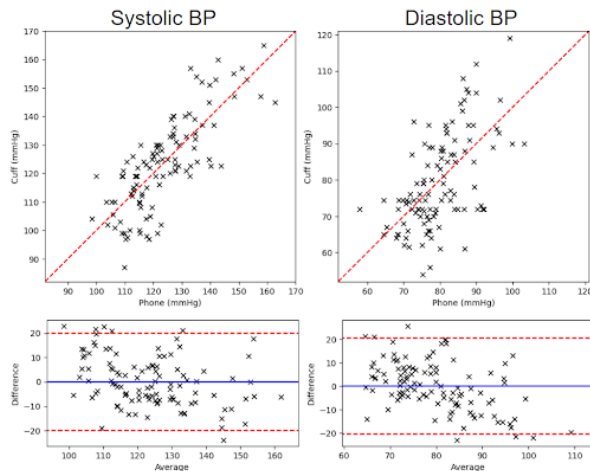


Figure 2: Blood Pressure Feasibility Study Regression and Bland Altman Plots of Results

patients can provide highly accessible, low-cost and low burden improvements for health interventions.

4.1 Performance evaluations

Addressing the first hypothesis will require evaluating the performance of the smartphone device in comparison to clinical gold standards. Following a feasibility studies outlined below, the next step is to make subsequent improvements and test the systems following clinical standards.

For BP, previous results from initial studies shown in Figure 2 suggest strong potential to improve the accuracy to clinically acceptable levels. As such, the future study evaluation will involve pursuing FDA approval through ISO 81060-2. The study protocol within the ISO requires an 85 person study with specific diversity distribution of BP, ethnicity, age, BMI, etc. Importantly, the protocol requires the ground truth BP measurements to be performed by 2 trained physicians using the gold standard auscultation method. The goal of this evaluation would be to meet the international standards for clinically suitable BP measurements.

The HGS evaluation will be performed in an N=85 person study, with comparisons to two high accuracy dynamometers. The study recruitment will ensure sufficient range and distribution of participants across demographics and HGS levels. Prior study results shown in Figure 3 indicates functionality across smartphones.

4.2 Deployment evaluations

The purpose of the deployment is the test the functionality of the system of improving health management for real users. The deployment will involve a training and interaction phase. The training will determine to what extent users are capable of using the system with only low-burden, remote training from the smartphone. Partnering with UC San Diego health, we will provide N=35 presurgical patients with the smartphone application. The application will prompt the users to perform exercises, activities, or lifestyle changes over the course of 1 month to improve their presurgical health state as measured by their BP and HGS.

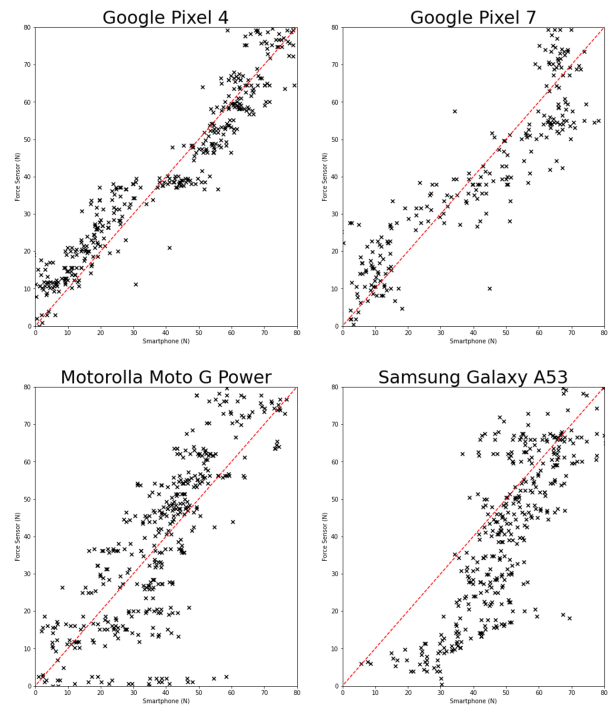


Figure 3: Regression and Bland Altman Plots of each Smartphone Performance in estimating applied force from hand grip strength

5 External endeavors

Beyond the academic pursuits of demonstrating the potential for functionality and impact, I intend to build a functional product that can be provide individuals with access to this technology. As such, I co-founded a company, Billion Labs Inc, with the goal of making clinically relevant health metrics widely available to smartphone users. Within the company, we have received indirect from NIH through the A2Collective foundation and are currently working with Element Boulder, a contract research organization that specializes in performing FDA validation studies.

6 Expected Contribution

The expected contribution is not only to develop novel measurement techniques for clinical measurements on smartphones, but also to insist on a new standard for health research. My hope is to push the field towards more interdisciplinary research that requires investigators to work closely with health professionals and think more about "What is important to measure?" rather than "What can we measure?".

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